

Immunodeficiency 88 (IMD88): Human T-bet Deficiency — Comprehensive Disease Characteristics Report

Summary

Immunodeficiency 88 (IMD88; OMIM #619630; MONDO:0030483; MedGen C5562026) is an ultra-rare, autosomal recessive **inborn error of IFN- γ immunity** caused by complete loss-of-function of the T-box transcription factor **T-bet**, encoded by **TBX21** (17q21.32; HGNC:11599; NCBI Gene 30009; UniProt O95936). It was first defined in a single consanguineous Moroccan boy who was homozygous for the in-frame indel **TBX21 c.466_471delGAGATGinsAGTTTA (p.E156_M157delinsSerLeu)**, a two-residue substitution within the T-box DNA-binding domain that abolishes DNA binding, impairs protein production and nuclear translocation, and prevents induction of *IFNG* ([PMID: 33296702](#)).

The disease is mechanistically remarkable because **one genetic lesion produces two distinct clinical arms**. First, loss of T-bet abolishes the development and IFN- γ output of **innate and innate-like lymphocytes** — natural killer (NK) cells, invariant NKT (iNKT), mucosal-associated invariant T (MAIT), and V δ 2⁺ $\gamma\delta$ T cells — placing IMD88 firmly within the **Mendelian Susceptibility to Mycobacterial Disease (MSMD)** spectrum and producing disseminated *bacille Calmette-Guérin* (BCG) disease. Second, loss of T-bet-mediated repression of the **Th2 program** derepresses IL-4/IL-5/IL-9/IL-13 production, driving blood **eosinophilia** and persistent **upper-airway/asthma-like inflammation** ([PMID: 33296702](#); [PMID: 34160550](#)).

Because IMD88 has been reported in only a single patient, most disease-level characteristics (epidemiology, prognosis, treatment algorithms) are **extrapolated** from the broader MSMD framework and from the well-characterized *Tbx21*-knockout mouse, which spontaneously recapitulates the allergic-airway arm of the human disease ([PMID: 11786643](#)). This report organizes all available evidence across the 15 requested domains, flags where information is inferred rather than demonstrated, and supplies ontology term suggestions (HPO, GO, CL, UBERON, NCIT, MONDO) throughout to support knowledge-base curation.

Key Findings

Finding 1 — IMD88 is autosomal-recessive T-bet (TBX21) deficiency causing mycobacterial disease

IMD88 is catalogued as OMIM #619630, MONDO:0030483, and MedGen C5562026, and is caused by homozygous mutation in **TBX21** (T-box transcription factor 21; T-bet; OMIM 604895) located at chromosome **17q21.32**. The disorder was defined in a single index patient — a boy born to **consanguineous Moroccan parents** — reported by Yang et al. in *Cell* (2020). Whole-exome sequencing combined with genome-wide linkage identified a homozygous deletion/insertion in *TBX21*. In vitro reconstitution in HEK293T cells demonstrated that the mutant allele produces protein poorly, translocates to the nucleus poorly, fails to bind target regulatory DNA elements, and fails to induce *IFNG* transcription. The authors state directly: "*We report a patient with mycobacterial disease due to inherited deficiency of the transcription factor T-bet*" (PMID: 33296702). This establishes IMD88 as a monogenic, recessive loss-of-function disorder of a master immune transcription factor.

Finding 2 — T-bet deficiency abolishes innate/innate-like IFN- γ -producing lymphocytes

The cellular basis of the mycobacterial susceptibility is a selective failure of the IFN- γ -producing innate compartment. The patient had **extremely low circulating counts of Mycobacterium-reactive NK, iNKT, MAIT, and V δ 2⁺ $\gamma\delta$ T lymphocytes**, along with reduced classical Th1 cells; the residual cells produced abnormally little IFN- γ . Notably, CD8⁺ $\alpha\beta$ T cells and non-classical CD4⁺ $\alpha\beta$ TH1 cells produced IFN- γ normally in response to mycobacterial antigens but could not compensate for the missing innate-like output. The authors summarize: "Human T-bet deficiency thus underlies mycobacterial disease by preventing the development of innate (NK) and innate-like adaptive lymphocytes (iNKT, MAIT, and V δ 2⁺ $\gamma\delta$ T cells) and IFN- γ production by them"* (PMID: 33296702). This identifies the specific developmental and functional lesion (loss of a discrete IFN- γ -producing cellular compartment) as the proximate cause of disease.

Finding 3 — Clinical phenotype: disseminated BCG, asthma/reactive airway disease, eosinophilia

The HPO/MedGen clinical profile (MedGen C5562026) includes **BCGosis / disseminated BCG infection (HP:0032262)**, **Asthma (HP:0002099)**, **Eosinophilia / increased eosinophil count (HP:0001880)**, and a general **Abnormality of the immune system (HP:0002715)**. OMIM #619630 notes **persistent reactive airway disease** associated with increased Th2 cytokine production and decreased IFN- γ . Despite serologic evidence of exposure to numerous viruses and bacteria, the patient did **not** develop other clinical infectious diseases, indicating a relatively selective, mycobacteria-dominant infection phenotype layered on top of an allergic/atopic airway disease ([PMID: 33296702](#)).

Finding 4 — T-bet loss derepresses the Th2 program (the second mechanistic arm)

A follow-up study (Yang et al., *J Exp Med* 2021) established the mechanism of the allergic arm. The patient's mutant T-bet **failed to inhibit Th2 cytokine production (IL-4, IL-5, IL-9, IL-13)** when overexpressed in Th2 cells; Herpesvirus saimiri (HVS)-immortalized patient T cells overproduced Th2 cytokines; **plasma IL-5 and IL-13 were markedly elevated**; and patient CD4⁺ $\alpha\beta$ T cells produced excess Th2 cytokines upon chronic stimulation **regardless of antigen specificity**, an effect reversed by wild-type T-bet. The result is **blood eosinophilia and persistent upper airway inflammation (UAI)**. The authors state: "*T-bet deficiency thus underlies the excessive production of Th2 cytokines, particularly IL-5 and IL-13, by CD4⁺ $\alpha\beta$ T cells, causing blood eosinophilia and UAI*" ([PMID: 34160550](#)). This confirms that the atopic manifestations are a direct consequence of the same TBX21 lesion, not a coincidental comorbidity.

Finding 5 — The Tbx21-knockout mouse recapitulates the asthma/airway arm

Finotto et al. (*Science* 2002) showed that **mice with targeted deletion of the *Tbx21* (T-bet) gene**, and SCID mice reconstituted with CD4⁺ cells from T-bet-knockout mice, **spontaneously developed multiple physiological and inflammatory features characteristic of asthma in the absence of allergen exposure**. Human asthmatic airway T cells showed reduced T-bet expression. The authors report: "*Mice with a targeted deletion of the T-bet gene and severe combined immunodeficient mice receiving CD4⁺ cells from T-bet knockout mice spontaneously*

demonstrated multiple physiological and inflammatory features characteristic of asthma" (PMID: 11786643). This provides a validated animal model for the allergic-airway component of IMD88, predating the human disease description by nearly two decades.

Finding 6 — IMD88 within the MSMD framework: diagnosis, treatment, prognosis

MSMD is defined by inborn errors of IFN- γ immunity, rendering patients *"highly and selectively susceptible to weakly virulent mycobacteria, such as environmental mycobacteria and Bacillus Calmette-Guérin vaccines"* (PMID: 32025907). A systematic review of **830 MSMD patients** (PMID: 38341181) reported a mean age of ~10.4 years, 52.5% male, a positive family history in 45.5%, highest reported frequencies in Iran/Turkey/Saudi Arabia, and 299 unique mutations across 21 genes. **Lymphadenopathy was the most common manifestation (45.5%)**, followed by fever (30.2%), organomegaly (24.8%), and sepsis (20.8%); *"Lymphadenopathy was the most common clinical manifestation of MSMD, reported in 378 (45.5%) cases."* MSMD carries substantial mortality, driven mostly by impaired control of infection. IMD88 (TBX21/T-bet deficiency) is one of these genetic etiologies, and its clinical management, diagnostic approach, and prognosis are reasonably extrapolated from this larger cohort.

Finding 7 — Exact TBX21 variant and gene/protein annotations

The index patient is **homozygous for TBX21 c.466_471delGAGATGinsAGTTTA**, an **in-frame insertion/deletion in exon 1** that replaces two highly conserved amino acids, **E156 and M157 (p.E156_M157delinsSerLeu)**, within the **T-box DNA-binding domain**. Parents were heterozygous carriers (WT/M); the patient was homozygous (M/M). The variant is **private** (not a recurrent/founder allele; absent as a benign homozygote in gnomAD). Gene/protein annotations: human **TBX21** (HGNC:11599; NCBI Gene 30009; UniProt O95936; 17q21.32); mouse ortholog **Tbx21** (NCBI Gene 57765; MGI:1888984; chromosome 11). The functional class is **complete autosomal-recessive loss of function** — the mutant protein shows impaired production, impaired nuclear translocation, and abolished DNA binding, failing to induce IFN- γ (PMID: 33296702).

Report by Requested Domain

1. Disease Information

- **Overview:** IMD88 is a monogenic inborn error of immunity in which loss of the master Th1/innate transcription factor T-bet produces a combined phenotype of **mycobacterial susceptibility** (MSMD-spectrum) and **allergic airway disease with eosinophilia**.
- **Key identifiers:** OMIM #619630; MONDO :0030483; MedGen C5562026; gene OMIM 604895 (TBX21). No dedicated Orphanet or distinct ICD-10/ICD-11 code exists for this ultra-rare entity; it is best coded under broad immunodeficiency categories (e.g., ICD-10 **D84.9**, immunodeficiency unspecified; ICD-11 **4A00** primary immunodeficiencies) and, for the infectious arm, under atypical mycobacterial infection. MeSH lacks a specific descriptor; relevant MeSH concepts include *Mycobacterium Infections*, *Immunologic Deficiency Syndromes*, and *T-Box Domain Proteins*.
- **Synonyms / alternative names:** "Immunodeficiency 88"; **human T-bet deficiency**; **TBX21 deficiency**; "Mendelian susceptibility to mycobacterial disease due to T-bet deficiency."
- **Information source:** Disease-level aggregated resources (OMIM, MONDO, MedGen) built on **individual-patient** primary reports (Yang et al. 2020/2021) — i.e., a single-patient basis, not EHR-scale or registry data.

2. Etiology

- **Causal factor:** Purely **genetic** — biallelic loss-of-function in *TBX21*. Environmental exposure (**BCG vaccination**) acts as the trigger that unmasks the mycobacterial susceptibility; the vaccine strain itself becomes the disseminating pathogen.
- **Genetic risk factors:** The single causal genotype is homozygous **TBX21 c.466_471delGAGATGinsAGTTTA**. **Consanguinity** is a major risk enabler (parents were related, consistent with autosomal-recessive inheritance and homozygosity by descent). No susceptibility loci or modifier genes have been formally mapped in this ultra-rare disease.
- **Environmental risk factors:** Exposure to **live BCG vaccine** or environmental non-tuberculous mycobacteria is the key environmental trigger for the infectious arm. Allergen and airway-irritant exposures plausibly modulate the atopic arm (inferred from mouse/asthma biology).
- **Protective factors:** By direct inference, **avoidance of live mycobacterial vaccines** in genetically at-risk siblings is protective. No genetic protective/modifier alleles are described.

- **Gene–environment interaction:** The paradigmatic GxE is **TBX21 loss × BCG exposure → disseminated BCGosis**; without the environmental mycobacterial challenge the infectious phenotype may not manifest, whereas the Th2/eosinophilia arm appears to be cell-intrinsic and largely antigen-independent ([PMID: 34160550](#)).

3. Phenotypes

Phenotype	Type	HPO term	Onset	Severity	Frequency (n=1 + MSMD context)
Disseminated BCG infection (BCGosis)	Clinical sign / infection	HP:0032262	Childhood (post-vaccination)	Severe	Present in index patient; BCG complications in ~55% of BCG-vaccinated MSMD (PMID: 36630059)
Asthma / reactive airway disease	Clinical sign	HP:0002099	Childhood	Moderate, persistent	Present in index patient
Eosinophilia	Laboratory abnormality	HP:0001880	Childhood	Moderate–marked	Present in index patient
Elevated Th2 cytokines (IL-5, IL-13)	Laboratory abnormality	(no direct HPO)	Childhood	Marked	Present (PMID: 34160550)
Upper airway inflammation	Clinical sign	HP:0012384 (airway)	Childhood	Persistent	Present
Abnormality of the immune system	General	HP:0002715	Childhood	—	Present
Lymphadenopathy (MSMD-context)	Clinical sign	HP:0002716	Childhood	Variable	Most common MSMD feature (45.5%) (PMID: 38341181)

- **Progression:** The mycobacterial arm is **episodic/infection-driven**; the atopic arm is **chronic/persistent**.
- **Quality-of-life impact:** No disease-specific QoL instruments have been applied. By analogy, disseminated mycobacterial disease imposes major morbidity (hospitalization, prolonged

multidrug therapy) and chronic asthma reduces daily functioning; formal EQ-5D/SF-36 data are **not available**.

4. Genetic / Molecular Information

- **Causal gene:** **TBX21** (T-bet), 17q21.32; gene OMIM 604895; HGNC:11599; NCBI Gene 30009; UniProt O95936.
- **Pathogenic variant:** **c.466_471delGAGATGinsAGTTTA (p.E156_M157delinsSerLeu)**, in-frame indel in exon 1, within the T-box DNA-binding domain. **Variant type:** in-frame delins (structural at protein level, altering two conserved residues). **Classification:** Pathogenic by ACMG-style functional evidence (abolished DNA binding, failure to induce IFN- γ , segregation with recessive inheritance, absence in gnomAD). **Allele frequency:** private; absent as benign homozygote in gnomAD. **Origin:** germline. **Functional consequence:** complete **loss of function** (not gain-of-function or dominant-negative in the heterozygous state — parents unaffected).
- **Modifier genes / epigenetics / chromosomal abnormalities:** None described for IMD88 specifically. Relevant epigenetic biology: T-bet normally shapes the chromatin landscape of the *IFNG* locus and represses Th2 loci; the Th17→Th1 plasticity literature notes extensive epigenetic priming of *IFNG* controlled by T-bet-family factors ([PMID: 29275836](#)). No large-scale cytogenetic abnormalities are involved.

5. Environmental Information

- **Environmental factors:** Live attenuated **BCG vaccine** and environmental non-tuberculous mycobacteria are the operative environmental agents.
- **Lifestyle factors:** Not characterized; allergen exposure may aggravate the airway phenotype (inferred).
- **Infectious agents:** *Mycobacterium bovis* **BCG** (disseminated disease in the index patient) and, by MSMD analogy, environmental mycobacteria, *M. tuberculosis*, and *Salmonella* species ([PMID: 32025907](#); [PMID: 36630059](#)).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Homozygous **TBX21 c.466_471delinsAGTTTA (p.E156_M157delinsSerLeu)** leads to a T-bet protein with two altered residues in the T-box DNA-binding domain (demonstrated).

2. This **results in** impaired protein production, impaired nuclear translocation, and **abolished binding to target DNA regulatory elements** (demonstrated in HEK293T reconstitution) ([PMID: 33296702](#)).

3. Loss of T-bet transcriptional activity **branches** into two arms:

Arm A (immunodeficiency / mycobacterial disease): - 4a. Failure to activate the T-bet-dependent Th1/innate transcriptional program **leads to** failed development and maturation of NK, iNKT, MAIT, and V δ 2⁺ $\gamma\delta$ T lymphocytes (demonstrated: extremely low counts). - 5a. This **results in** loss of early, innate IFN- γ **production** against mycobacteria (demonstrated). - 6a. Deficient IFN- γ **leads to** impaired macrophage activation and failure to control weakly virulent mycobacteria (inferred from the MSMD paradigm) ([PMID: 32025907](#)). - 7a. This **results in disseminated BCG disease** and MSMD-spectrum susceptibility (demonstrated clinically).

Arm B (allergic airway disease / eosinophilia): - 4b. Loss of T-bet-mediated **repression of the Th2 program leads to** derepressed IL-4/IL-5/IL-9/IL-13 output by CD4⁺ $\alpha\beta$ T cells, independent of antigen specificity (demonstrated) ([PMID: 34160550](#)). - 5b. Elevated **IL-5 results in blood eosinophilia**; elevated **IL-13 results in** airway mucus/inflammation (demonstrated: elevated plasma IL-5/IL-13). - 6b. Sustained type-2 inflammation **leads to persistent upper-airway inflammation and asthma/reactive airway disease** (demonstrated clinically; recapitulated in *Tbx21*-KO mice, [PMID: 11786643](#)).

- **Molecular pathways:** IFN- γ -STAT1-T-bet axis (upstream), IL-12/IL-12R β 1 signaling context of MSMD, and the Th1/Th2 master-regulator circuit (T-bet vs GATA3). TREM-2 has been shown to feed into T-bet induction via the CD3 ζ -ZAP70/IFN- γ R-STAT1/STAT4 route in the context of *M. tuberculosis* ([PMID: 34623322](#)).
- **Cellular processes:** Lymphocyte lineage commitment/differentiation; type-1 vs type-2 immune polarization; macrophage activation; inflammation.
- **Protein dysfunction:** DNA-binding loss of function of a transcription factor (not misfolding/aggregation).
- **Immune involvement:** Combined **immunodeficiency** (innate IFN- γ arm) and **immune dysregulation/allergy** (Th2 arm) — a rare "two diseases, one gene" configuration.
- **Suggested GO terms:** GO:0045063 (T-helper 1 cell differentiation), GO:0045064 (T-helper 2 cell differentiation), GO:0032609 (interferon-gamma production), GO:0003700 (DNA-binding transcription factor activity), GO:0006357 (regulation of transcription by RNA Pol II).
- **Suggested CL terms:** CL:0000623 (NK cell), CL:0000814 (mature NK T cell / iNKT), CL:0000940 (mucosal invariant T cell / MAIT), CL:0000798 (gamma-delta T cell), CL:0000546 (T-helper 2 cell), CL:0000545 (T-helper 1 cell), CL:0000771 (eosinophil).

7. Anatomical Structures Affected

- **Organ/system level:** **Immune/lymphoid system** (primary); **respiratory system** — airways/lungs (UBERON:0001004 respiratory system; UBERON:0002048 lung; UBERON:0001005 respiratory airway); **upper airway / nasal mucosa** (UBERON:0001728 nasopharynx region). Secondary involvement in disseminated BCG can affect **lymph nodes** (UBERON:0000029), **spleen/liver** (organomegaly in MSMD context), **skin**, and **bone**.
- **Tissue/cell level:** Airway mucosal epithelium and submucosal inflammatory infiltrate (eosinophils, Th2 cells); lymphoid tissue with deficient innate-like lymphocytes. Cell populations: NK, iNKT, MAIT, V δ 2⁺ $\gamma\delta$ T (deficient); Th2 cells and eosinophils (expanded).
- **Subcellular level:** **Nucleus** (GO:0005634) — the site of the primary transcription-factor defect (impaired nuclear translocation and DNA binding).
- **Localization/lateralization:** Airway disease is **bilateral/diffuse**; disseminated BCG is **systemic**.

8. Temporal Development

- **Onset:** **Childhood**, typically after BCG vaccination for the infectious arm; airway/atopic features also present in childhood. Onset is **subacute-to-chronic**.
- **Progression:** Infectious episodes are **episodic/relapsing** with treatment; airway disease is **chronic and persistent**. Disease is **lifelong** (germline monogenic).
- **Critical period:** The **peri-vaccination window** (neonatal BCG in endemic regions) is the key period of vulnerability and the key opportunity for prevention (avoiding live BCG in at-risk siblings).

9. Inheritance and Population

- **Inheritance:** **Autosomal recessive** (biallelic loss of function; heterozygous parents unaffected).
- **Penetrance/expressivity:** Cannot be estimated from n=1; MSMD generally shows incomplete penetrance and variable expressivity across genes.
- **Epidemiology:** **Ultra-rare** — one reported patient worldwide. No prevalence/incidence figures exist for IMD88 specifically. MSMD as a class: mean age at presentation ~10.4 yr, 52.5% male, positive family history 45.5%, clustering in Iran/Turkey/Saudi Arabia and consanguineous populations ([PMID: 38341181](#)); in Morocco, MSMD comprised ~50% of genetically confirmed innate/intrinsic IEI ([PMID: 41209815](#)), and *TBX21* accounted for 1 of 22 MSMD patients across 15 Moroccan kindreds ([PMID: 36630059](#)).

- **Consanguinity/founder:** **Consanguinity** is central to case ascertainment; the variant is **private/non-founder**.
- **Carrier frequency:** Not established; expected extremely low given the private allele.

10. Diagnostics

- **Immunologic laboratory tests:** Flow-cytometric enumeration of **NK, iNKT, MAIT, and Vδ2⁺ γδ T cells** (markedly reduced); functional **IFN-γ production assays** (BCG ± IL-12 stimulation of whole blood), which are reduced ([PMID: 33296702](#); [PMID: 36630059](#)). **Blood eosinophil count** (elevated; LOINC 26449-9) and **plasma IL-5/IL-13** (elevated) support the atopic arm.
- **Biomarkers:** Low innate IFN-γ output; elevated Th2 cytokines (IL-5, IL-13); eosinophilia.
- **Genetic testing (definitive):** **Whole-exome sequencing** (as used in the index case) or **whole-genome sequencing**, ideally combined with linkage/homozygosity mapping in consanguineous families; targeted **TBX21** single-gene/panel testing once suspected. Upfront genomic sequencing is increasingly recommended for primary atopic disorders and IEI with red-flag features ([PMID: 39381601](#)). CMA/karyotype/FISH/mtDNA/repeat testing are **not applicable**.
- **Clinical criteria/differential:** Diagnosis rests on the **MSMD phenotype** (susceptibility to weakly virulent mycobacteria) plus molecular confirmation. Differential diagnosis: other MSMD genes (**IL12RB1, IL12B, IFNGR1/2, STAT1, ISG15, IRF8, SPPL2A, TYK2, RORC, JAK1, CYBB, NEMO**) ([PMID: 30264912](#)); and, for the atopic arm, other **primary atopic disorders** and hyper-eosinophilic syndromes.
- **Screening:** In BCG-endemic, consanguineous families, **cascade genetic screening** of siblings and **deferral of live BCG** until MSMD is excluded is advised ([PMID: 36630059](#)).

11. Outcome / Prognosis

- **Survival/mortality:** No IMD88-specific survival data (n=1). MSMD as a class carries **substantial mortality driven by uncontrolled infection**; prognosis depends on the specific genetic etiology and infection control ([PMID: 38341181](#)).
- **Morbidity:** Recurrent/disseminated mycobacterial infection and chronic asthma/eosinophilic airway disease. The index patient did not develop other clinical infections despite broad serologic exposure, suggesting a relatively selective infection risk.

- **Prognostic factors (inferred):** Timeliness of antimycobacterial therapy, avoidance of further live-vaccine exposure, degree of residual IFN- γ immunity, and access to HSCT would be expected to modify outcome.

12. Treatment

(All treatment is extrapolated from MSMD and Th2-directed asthma care; no IMD88-specific trials exist.)

- **Antimycobacterial pharmacotherapy:** Prolonged **multidrug antimycobacterial regimens** (e.g., rifampin, isoniazid, ethambutol, a macrolide/fluoroquinolone as appropriate) for disseminated BCG/mycobacterial disease (NCIT: antimycobacterial/antitubercular agents).
- **Immunomodulation: Recombinant IFN- γ 1b** (NCIT:C1732, Interferon Gamma-1b) as adjunctive therapy in IFN- γ -pathway MSMD; rationale is to bolster deficient IFN- γ signaling, though efficacy in a downstream transcription-factor defect that impairs IFN- γ *production* and responsiveness is uncertain.
- **Definitive therapy: Hematopoietic stem cell transplantation (HSCT)** (NCIT:C15431) is potentially curative for severe combined IFN- γ -immunity defects; candidacy must be individualized.
- **Anti-Th2 / asthma therapy:** Standard asthma control (inhaled corticosteroids, bronchodilators) plus **biologics targeting the type-2 axis** — anti-IL-5/anti-IL-5R α (mepolizumab/benralizumab), anti-IL-4R α (dupilumab), or anti-IgE (omalizumab) — are mechanistically rational given elevated IL-5/IL-13 and eosinophilia ([PMID: 34160550](#)).
- **Preventive: Avoid live BCG** and other live vaccines.
- **Pharmacogenomics/experimental:** None specific; **gene therapy/gene editing of TBX21** is conceptual only.

13. Prevention

- **Primary prevention:** In affected kindreds, **withhold/defer live BCG vaccination** in newborn siblings until MSMD is genetically excluded ([PMID: 36630059](#)).
- **Secondary prevention: Cascade genetic screening** and early flow-cytometric/IFN- γ functional testing of at-risk relatives; early recognition of mycobacterial disease.
- **Tertiary prevention:** Prompt, adequate antimycobacterial therapy and type-2-directed asthma control to prevent complications.

- **Genetic counseling:** Recessive inheritance with 25% sibling recurrence risk; **prenatal/preimplantation testing** possible once the familial variant is known. Consanguinity counseling is relevant.

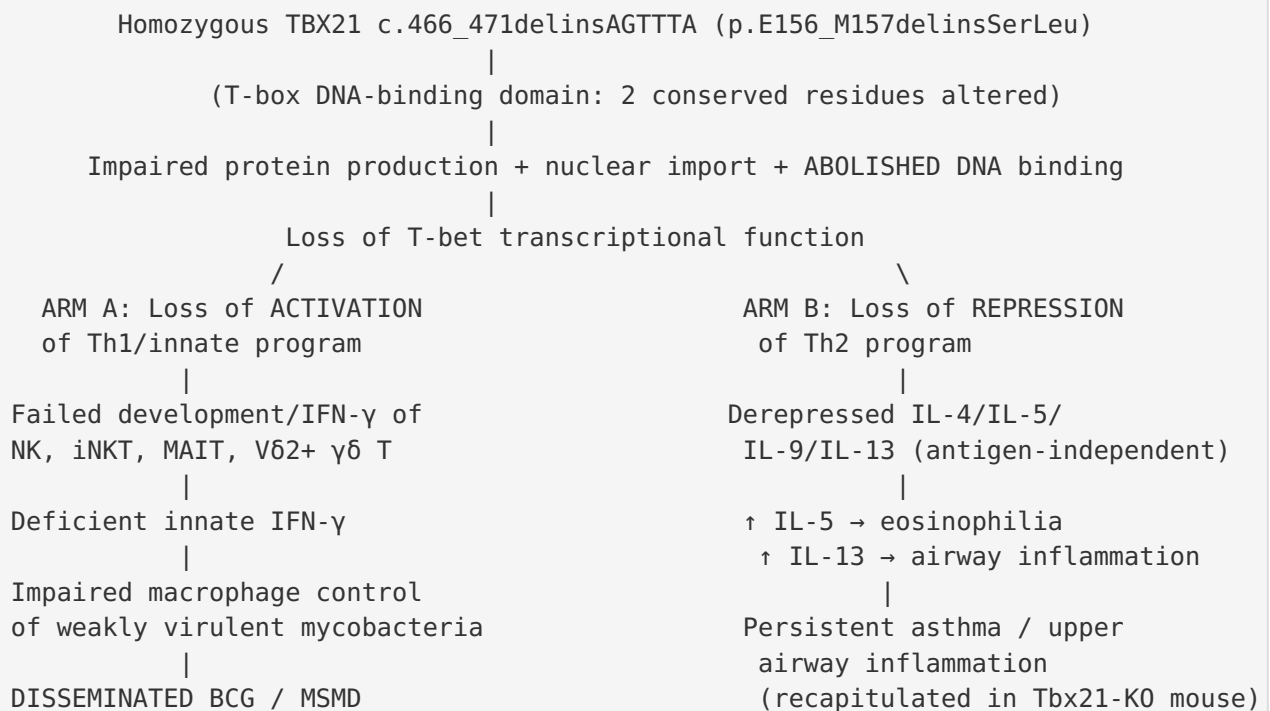
14. Other Species / Natural Disease

- **Taxonomy/orthologs:** Mouse **Tbx21** (NCBI Gene 57765; MGI:1888984; chromosome 11) is the direct ortholog of human **TBX21**. T-bet is evolutionarily conserved as the master Th1/type-1 regulator across mammals; ILC1s and Th1 cells across species depend on it ([PMID: 35163778](#); [PMID: 33126494](#)).
- **Natural disease in other species:** No spontaneously occurring companion-animal or wildlife equivalent of IMD88 has been reported (no dedicated OMIA entry). Disease knowledge is confined to engineered mouse models.
- **Zoonotic/transmission:** Not applicable — IMD88 is a non-transmissible germline disorder.

15. Model Organisms

- **Primary model: *Tbx21*/T-bet knockout mouse** — spontaneously develops asthma-like airway physiology and inflammation without allergen challenge, and SCID mice reconstituted with T-bet-KO CD4⁺ cells reproduce the phenotype ([PMID: 11786643](#)).
 - **Model types available:** Constitutive knockout; adoptive-transfer/reconstitution models; conditional/CD4-specific approaches used in related TREM-2/T-bet work ([PMID: 34623322](#)).
 - **Phenotype recapitulation:** The mouse KO **strongly recapitulates the allergic-airway arm** (Th2 derepression, airway inflammation). The mycobacterial-susceptibility arm is supported by the general dependence of Th1/innate IFN- γ immunity on T-bet, but the human innate/innate-like lymphocyte deficiency (iNKT/MAIT/V δ 2⁺) is only partially mirrored in mice, whose innate-like compartments differ.
 - **In-vitro models:** HEK293T reconstitution assays (used to prove the DNA-binding/IFN- γ -induction defect) and HVS-immortalized patient T-cell lines (used to prove Th2 overproduction) ([PMID: 33296702](#); [PMID: 34160550](#)).
 - **Resources:** MGI (*Tbx21*, MGI:1888984); IMPC/KOMP for T-bet alleles.
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Mechanistic Model / Interpretation



The unifying insight is that **T-bet is simultaneously an activator of type-1 immunity and a repressor of type-2 immunity**. A single loss-of-function lesion therefore removes both functions, yielding the paradoxical combination of an **immunodeficiency** (susceptibility to mycobacteria) and an **immune-dysregulation/allergy** phenotype (eosinophilic airway disease) in the same patient. Arm A is the demonstrated cause of the MSMD phenotype and is developmental (missing cell lineages); Arm B is cell-intrinsic, antigen-independent, and directly reversible by wild-type T-bet in vitro. The mouse model validates Arm B decisively and supports Arm A through the conserved T-bet dependence of Th1/IFN-γ immunity.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
33296702	<i>Human T-bet Governs Innate and Innate-like Adaptive IFN-γ Immunity against Mycobacteria</i>	Human clinical + in vitro	Defines IMD88; identifies variant; proves DNA-binding/IFN- γ defect; establishes Arm A cellular mechanism
34160550	<i>High Th2 cytokine levels and upper airway inflammation in human inherited T-bet deficiency</i>	Human clinical + in vitro	Proves Arm B: Th2 derepression $\rightarrow$ IL-5/IL-13 $\rightarrow$ eosinophilia/UAI
11786643	<i>T-bet KO mice spontaneously develop asthma features</i>	Mouse model	Validates the allergic-airway arm; provides model organism
32025907	<i>MSMD: recent discoveries</i>	Review	Defines MSMD disease class to which IMD88 belongs
38341181	<i>830 MSMD patients: systematic review</i>	Human cohort	Epidemiologic/clinical context, prognosis
36630059	<i>MSMD in 22 Moroccan patients (incl. 1 TBX21)</i>	Human cohort	Population context; BCG-complication rate; TBX21 in Moroccan cohort
41209815	<i>Innate/intrinsic immunity in Morocco</i>	Human registry	MSMD prevalence in Moroccan IEI registry
30264912	<i>MSMD: 2014–2018 update</i>	Review	Differential diagnosis / MSMD gene list
34623322	<i>TREM-2 promotes Th1 via CD3ζ-ZAP70</i>	Mouse/human mechanistic	Upstream signaling context for T-bet induction
39381601	<i>Rapid identification of primary atopic disorders by genomic sequencing</i>	Review	Diagnostic strategy (upfront WGS)
35163778 / 33126494	<i>ILCs / CD4 Th subsets reviews</i>	Review	T-bet/GATA3 master-regulator biology; conservation
29275836	<i>Th17 plasticity / IFNG epigenetics</i>	Review	Epigenetic context of T-bet/IFNG regulation

Limitations and Knowledge Gaps

1. **Single-patient basis (n=1).** Every disease-specific characteristic — penetrance, expressivity, full phenotype spectrum, natural history, treatment response, prognosis — rests on one individual. Population-level parameters are borrowed from the broader MSMD class and must be interpreted as inference, not established fact for IMD88.
 2. **No treatment evidence.** No therapy has been tested specifically in T-bet deficiency. IFN- γ 1b efficacy is uncertain because the defect lies downstream at a transcription factor that impairs both IFN- γ production and responsiveness. Type-2 biologics (anti-IL-5/IL-4R α) are mechanistically rational but untested in this disorder.
 3. **Incomplete model concordance.** The mouse *Tbx21*-KO robustly models the airway/Th2 arm but only partially models the human innate/innate-like lymphocyte deficiency, because murine iNKT/MAIT/ $\gamma\delta$ compartments differ from human.
 4. **No epidemiology, QoL, imaging, or omics datasets** exist for IMD88 as a distinct entity; ICD coding and Orphanet classification are not specifically assigned.
 5. **Modifier genetics and penetrance** are entirely unknown; whether heterozygous carriers have subtle immune/atopic phenotypes has not been examined.
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Proposed Follow-up Experiments / Actions

1. **Case-finding / cohort expansion:** Systematically screen MSMD and unexplained eosinophilia/severe-asthma cohorts (especially consanguineous, BCG-endemic populations) for biallelic *TBX21* variants via WES/WGS and GeneMatcher, to move IMD88 beyond n=1 and define its true phenotype spectrum and penetrance.
2. **Therapeutic proof-of-concept:** Evaluate **type-2-targeting biologics (mepolizumab/benralizumab/dupilumab)** for the eosinophilic-airway arm and formally assess **HSCT** outcomes and **IFN- γ 1b** response for the mycobacterial arm in any newly identified patients.
3. **Single-cell/transcriptomic profiling:** Perform scRNA-seq/ATAC-seq on patient PBMCs to map, at cellular resolution, the loss of innate-like lymphocytes and the derepressed Th2 program, and to identify T-bet direct target loci that fail activation vs fail repression.
4. **Structure-function studies:** Model p.E156_M157delinsSerLeu in the T-box domain (crystallography/AlphaFold + EMSA) to define precisely how the two-residue substitution abolishes DNA binding, informing potential correction strategies.

5. **Refined animal models:** Generate **humanized or knock-in mice carrying the patient allele**, and models that better reconstitute human innate-like lymphocyte compartments, to test the mycobacterial arm and candidate therapies.
 6. **Carrier / heterozygote studies:** Assess immune and atopic phenotypes in obligate heterozygous relatives to determine whether *TBX21* haploinsufficiency contributes to common asthma/atopy risk.
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Report compiled from 5 investigation iterations, 7 confirmed findings, and 22 reviewed papers. Evidence types are distinguished throughout as human clinical, model organism, in vitro, or review/inference. Where information is unavailable for IMD88 specifically, this is stated explicitly and MSMD-class data are used as the nearest available proxy.

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